A Fully Automated Method for Lung Nodule Detection From Postero-Anterior Chest Radiographs

Paola Campadelli, *Member, IEEE*, Elena Casiraghi*, and Diana Artioli

***Abstract*—In the past decades, a great deal of research work has been devoted to the development of systems that could improve radiologists' accuracy in detecting lung nodules. Despite the great efforts, the problem is still open. In this paper, we present a fully automated system processing digital postero-anterior (PA) chest radiographs, that starts by producing an accurate segmentation of the lung field area. The segmented lung area includes even those parts of the lungs hidden behind the heart, the spine, and the diaphragm, which are usually excluded from the methods presented in the literature. This decision is motivated by the fact that lung nodules may be found also in these areas. The segmented area is processed with a simple multiscale method that enhances the visibility of the nodules, and an extraction scheme is then applied to select potential nodules. To reduce the high number of false positives extracted, cost-sensitive support vector machines (SVMs) are trained to recognize the true nodules. Different learning experiments were performed on two different data sets, created by means of feature selection, and employing Gaussian and polynomial SVMs trained with different parameters; the results are reported and compared. With the best SVM models, we obtain about 1.5 false positives per image (fp/image) when sensitivity is approximately equal to 0.71; this number increases to about 2.5 and 4 fp/image when sensitivity is ≈ 0.78 and ≈ 0.85, respectively. For the highest sensitivity (≈ 0.92 and 1.0), we get 7 or 8 fp/image.**

***Index Terms*—Feature selection, nodule detection, support vector machine (SVM) classification.**

I. Introduction

LUNG CANCER is a major cause of mortality in the western world, as demonstrated by the striking statistical numbers published every year by the American Lung Cancer Society. They indicate that the five year survival rate for patients with lung cancer can be improved from an average of 14% up to 49% if the disease is diagnosed and treated at its early stage [1], when these lesions are bright round-shaped abnormalities, generally referred as nodules. Several other studies, reported in [2]–[5], also proved that the survival rate can be improved greatly if the tumor is removed at this stage.

The most common techniques currently used to detect lung tumors include chest radiography, cytologic analysis of sputum samples, fiber-optic examination of bronchial airways, and, finally, computerized tomography scans (CT) and magnetic resonance imaging (MRI). Although the last two are more sensitive and precise, in many parts of the world widespread screening by CT or MRI is not yet practical, so that chest radiology remains the initial and most common procedure. This is due to its advantages: it is the most cost-effective, the most routinely available, the most dose-effective diagnostic tool, often preferred also for its ability to reveal some unsuspected pathologic alterations (see [1] and [6]). We may add that no nation currently has a screening program in place for lung cancer, so that early-stage lung cancer is often discovered in the form of solitary lung nodule when a patient is X-rayed for another purpose [7].

All these reasons explain why the detection of subtle or early-stage lung nodules on chest radiographs is one of the outstanding challenges in the field of medical diagnosis. Striking statistics about diagnosticians' error rates have been reported by several studies over the past four decades [8]–[12], and the fact that no improvement in these results has been observed even when employing more sophisticated imaging techniques [13], shows the potential for improving early diagnosis and suggests the use of computer programs for X-ray images analysis.

A computer approach is attractive because it has the potential to provide objective and consistent results; thus, an automatic scheme that alerts radiologists to the location of highly suspect lung nodules should allow the number of diagnostic errors to be reduced. This could lead to earlier detection of primary lung cancer and of metastatic nodules and, it is hoped, a better prognosis for the patient. Moreover, the advent of digital thorax units and digital radiology departments with picture archiving communication systems (PACS) makes it possible to use computerized methods to analyze chest radiographs on a routine basis. A successful detection scheme could eventually be hardwire implemented for online screening of all chest radiographs, prior to viewing by a physician. Thus, chest radiographs ordered for medical reasons other than suspected lung cancer would also undergo careful screening for nodule detection.

The feasibility of computer-aided diagnosis (CAD) systems for lung nodule detection in chest radiographs was demonstrated by observer performance studies reported by several authors (see [14]–[17]); they proved the feasibility of improving radiologists' detection accuracy by means of a system to provide radiologists with a "second opinion."

All these reasons motivate the wide variety of computerized systems aimed at lung-nodule detection, proposed in the last two decades (e.g., [14] and [18]–[46]). Most of them have been reported in an interesting survey by Van Ginneken [47] and are based on a common processing scheme. At first, the chest image is manually or automatically segmented in order to restrict the processing area of the subsequent detection algorithms. The fol-

Manuscript received January 12, 2006; revised July 31, 2006. This work was supported by CIMAINA and FIRST 2006 (University of Milan). *Asterisk indicates corresponding author.*

P. Campadelli is with Department of Computer Science, Università degli Studi di Milano, Milan 20135, Italy (e-mail: campadelli@dsi.unimi.it).

*E. Casiraghi is with Department of Computer Science, Universita degli Studi di Milano, Milan 20135, Italy (e-mail: casiragh@dsi.unimi.it).

D. Artioli is with the Radiology Department, Ospedale Niguarda Ca'Granda di Milano, Milan 20162, Italy (e-mail: diansa@alice.it).

Digital Object Identifier 10.1109/TMI.2006.884198

0278-0062/$20.00 © 2006 IEEE

lowing steps are aimed at the enhancement of the visibility of the nodules and at the extraction of an initial set of potential nodule regions. This is always characterized by high cardinality which makes the system useless for clinical purposes; a last step must then be added with the aim of reducing the number of false positives detected. To this end, classical and well-known classification techniques are employed, such as neural networks (NN) or K-nearest-neighbor (KNN) classifiers.

Although the widespread research work devoted to the development of lung-nodule detection systems, the problem is still open since none of the proposed methods can produce satisfactory results that make it practical for everyday screening procedures. Indeed, one method has been proposed in [17] and tested in a clinical environment to evaluate its effectiveness. As reported by the authors, "although the results of the observer study showed a significant improvement in the accuracy of lung nodule detection with CAD output images, the CAD performance level of 73% sensitivity is still not adequate to avoid overlooking lung nodules." The need is to further improve the sensitivity of lung-nodule detection algorithms. Another limitation, which is pointed out by the authors, is that this study was performed on images from a private database, making a comparison with other methods impossible. This is a general problem, common to most of the proposed techniques; a few of them (e.g., [44]) have been tested on X-ray images that belong to a standard database, while others use the standard database for training the system (e.g., [41]) and present their results on private images that contain nodules whose subtlety is not specified.

In this paper, we present our fully automated lung-nodule detection system applied to PA chest radiographs; it is based on the commonly used processing steps described above (segmentation, enhancement of nodule visibility, extraction of a nodule candidate set and classification). The method starts with lung field segmentation, whose result is a lung mask where even the areas hidden behind the heart, the spine, and the diaphragm are included; this is a different choice with respect to the lung segmentation methods presented in the literature, due to the fact that lung nodules may be found also in these areas. The segmented area is then processed by two different, consecutive multiscale schemes to enhance and extract the suspected nodules; they take into account all possible nodule sizes and the various contrast characteristics of nodules located in different lung areas.

The system has been tested on a standard database described in [48], which contains 154 radiographs of patients with lung nodules; the extracted nodule candidate set contains 151 nodules (i.e., a sensitivity equal to 0.98).[1]

Nevertheless, the cardinality of this set is quite high, since it contains 18916 false positive (fp) regions (approximately 77 fp per image), and must be reduced to make the system useful for clinical purposes. This is done by means of a feature selection technique, followed by classification of suspected nodules using different support vector machines (SVMs). With the best SVM models we obtain about 1.5 fp/image when sensitivity is approximately equal to 0.71; this number increases to about 2.5 and 4 fp/image when sensitivity is ≈ 0.78 and ≈ 0.85, respectively. For the highest sensitivities (≈ 0.92 and 1.0), we get 7 or 8 fp/image.

The structure of the paper is as follows. The segmentation algorithm is outlined in Section III. Section IV describes the multiscale scheme used to enhance the visibility of the nodules. Section V describes the candidate extraction step, which is an improved version of the method presented in [49] since it extracts a candidate set with a lower cardinality and a higher sensitivity. Section VI describes the experiments aimed at classifying potential nodules by means of SVMs, while Sections VII and VIII report the results obtained and the conclusions.

II. Materials

The whole method has been tested on a standard database acquired by the Japanese Society of Radiological Technology (JSRT), described in [48]. It contains 247 digital X-ray images: 154 containing lung nodules, and 93 of patients with no disease. The images have been digitized at a resolution of 0.165 mm pixel size. They have 2048×2048 pixels and 4096 gray levels. The diameter of the nodules ranges from 5 to 35 mm. All the nodules have been classified according to the difficulties encountered in their detection by the radiologists. They have been divided into five classes ranging from obvious to extremely subtle.

The segmentation algorithm was also tested on 240 radiographs acquired from the Niguarda Ca'Granda Hospital in Milan. They are different from the JSRT images both in resolution (0.160 mm pixel size) and in size (they have a maximum of 2128×2584 pixels, but this is a not fixed size).

III. Lung Field Segmentation

The first step in an automatic system for lung nodule detection, and for any further analysis of chest radiographs, is the segmentation of the lung field in order to restrict the processing area of the subsequent algorithms. The segmentation algorithms proposed in the literature and designed for different purposes are based on classical techniques (such as [26] and [50]–[62]). One of the limitations of these algorithms is that none of them considers the areas behind the heart, the spine, and the diaphragm (the hidden lung areas), where lung nodules may still be present. Moreover, they lack robustness since they are based on several assumptions, often violated in real-life situations, regarding chest position, size, and orientation. Other methods have been proposed (e.g., [63] and [64]), which get better results without making these strong assumptions, but simply by including some necessary knowledge about generic chest shape. Nevertheless, even these methods produce segmentation masks where hidden lung areas are excluded. Note that this is not necessarily a limitation, since the solution of certain medical tasks can be computed by restricting the processing to the visible lung area which contains all the required information; an example is the tuberculosis diagnosis, or the detection of other pathologies that, if present,

[1]The sensitivity is defined as $N_{\mathrm{TP}}/(N_{\mathrm{TP}} + N_{\mathrm{FN}})$, where N_{TP} is the number of true positives, i.e., nodules correctly detected, and N_{FN} is the number of false negative candidates, i.e., missed nodules. When assessing the performance of a classifier trained to recognize nodular and non-nodular regions, the specificity is also used; it is defined as $N_{\mathrm{TN}}/(N_{\mathrm{TN}} + N_{\mathrm{FP}})$, where N_{TN} is the number of true negatives, i.e., non-nodule regions correctly recognized, and N_{FP} is the number of false positives, i.e., non-nodule regions wrongly classified as nodules.

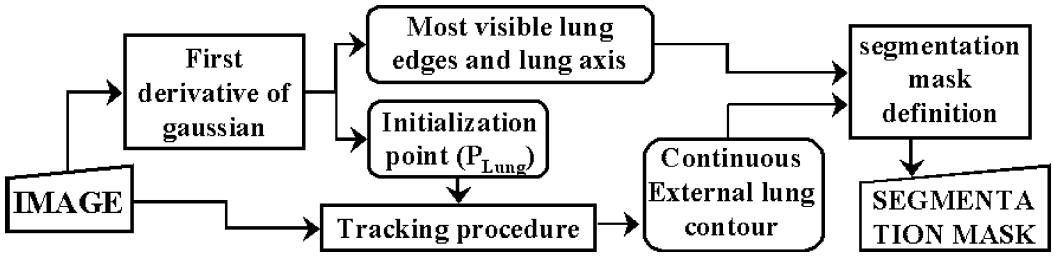


Fig. 1. Schematic diagram of the method used to segment the full lung area.

are restricted to the visible lung area or spread all over the lung field.

However, high precision is a strong requirement when the segmentation task is meant to be the first step in a lung nodule detection system, since a segmentation mask where (even small) parts are lost may miss some nodules. All these facts motivate the development of our segmentation method; it includes the hidden parts in the lung area too, and avoids assumptions regarding chest position, size, and orientation. It works with images where the chest is not always located in the central part of the image, may be tilted and may have structural abnormalities. In this paper, we only sketch the method, whose detailed description is reported in [65]. It works on images down-sampled to a dimension of 256×256 pixels; this size was chosen experimentally to reduce computation cost without worsening performance.

A precise external lung contour is obtained by properly combining the results of two simple image-segmentation techniques. Their appropriate integration takes generic chest shape into account and can therefore recognize and repair their complementary errors, by exploiting the good characteristics of their individual results. A schematic diagram of the method is shown in Fig. 1. The first algorithm detects the most visible lung edges by means of the first derivatives of Gaussian filters taken at four different orientations. The edges thus detected provide an initial outline of the lung borders. Moreover, they are used to define the vertical lung axis and to find a point, P_{Lung}, that surely belongs to the rib cage boundaries; this point is needed to initialize the second technique, which provides a continuous external lung contour. Indeed, P_{Lung} is the starting point for an edge-tracking procedure that works on three images representing the chest at three different levels of detail. The three representations are obtained by the application of the Laplacian of Gaussian (LoG) operator at three different scales.

Finally, the most visible lung edges and the continuous external lung contour, created by the two techniques, are integrated to define the external border of the segmentation mask. Simple techniques are then used to define the segmentation mask near the spine and to delimit it at the bottom. In the top row of Fig. 2, two results are shown.

Once the segmentation mask has been defined a further method has been developed to find the separation between the hidden and the visible lung areas. Since those two areas differ mainly in their pixel gray levels, the separation is performed by a contour following procedure whose result is integrated with the result produced by a gray-level clustering method [66]. Two

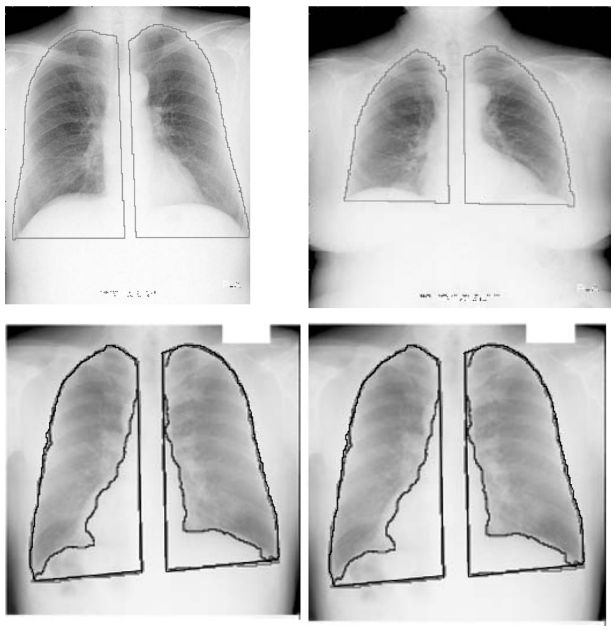


Fig. 2. Top row: Some of the results obtained by the full lung area segmentation algorithm. Bottom row: result of the separation between the hidden and the visible lung areas; the ground truth is black colored and superimposed on the images.

results are shown in the bottom row of Fig. 2. This separation step has been helpful in detecting and correcting some errors in the previously described segmentation algorithm. Furthermore, it has been used to increase the performance of our computerized lung-nodule detection system; indeed, the extraction step obtains better sensitivity by processing the two areas (hidden and visible) separately, and the classification step employs this information to better characterize candidate nodules.

We tested the segmentation algorithm both on the 247 images of the JSRT database and on the 240 radiographs from the Niguarda Ca'Granda Hospital. To judge the obtained results, we compared the segmentation masks, created by our algorithm, with the true lung masks (the ground truth) manually defined by two radiologists who independently traced the lung field borders.[2] The use of two observers should ensure an unbiased evaluation scheme. The true lung mask could be matched

[2]Note that, due to the inter-observer (radiologist) variability [64], the lung borders of the same patient traced independently by two radiologists, cannot be coincident, and they always present sparse nonmatching pixels. Although different, both the manually segmented lung masks are considered by radiologists equivalent and correct, and, for this reason, radiologists decided to create a unique true lung mask to be compared to the computer outlines.

to the segmented mask to classify each pixel in the segmented image as:

- true positive pixel (or TP): it is a pixels correctly included in the lung mask;
- false positive pixel (or FP): it is a pixel wrongly included in the segmented lung mask;
- true negative pixel (or TN): it is a pixel, belonging to the external structures or the background, correctly excluded by the segmented mask.
- false negative pixel (or FN): it is a pixel wrongly excluded by the segmented mask.

We could then evaluate, for each segmented image and for each lung separately, the following four performance measures[3]

$$\text{Positive predictive value} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \quad (1)$$

$$\text{Sensitivity} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \quad (2)$$

$$\text{Specificity} = \frac{N_{\text{TN}}}{N_{\text{FP}} + N_{\text{TN}}} \quad (3)$$

$$\text{Accuracy} = \frac{N_{\text{TP}} + N_{\text{TN}}}{N_{\text{TP}} + N_{\text{TN}} + N_{\text{FP}} + N_{\text{FN}}} \quad (4)$$

where N_{TP}, N_{FP}, N_{TN}, and N_{FN} are the number of the true positive, false positive, true negative, and false negative pixels, respectively. The same segmentation performance measures were used also by Van Ginneken in [64] and a similar approach was recently reported by Estrada in [67]. The accuracy measures the number of correctly classified pixels with respect to the total number of pixels in the image, but it does not make any distinction between types of error encountered (FP or FN). For this reason, the evaluation is mainly based on the positive predictive value and the sensitivity measurement: a low positive predictive value means that lots of background pixels were wrongly included in the lung mask, while low sensitivity is the result of major lung areas missing in the segmented mask.

On the overall set of 487 images in our database, the mean positive predictive value is equal to 0.9197, the mean sensitivity is equal to 0.9174, while the mean specificity and accuracy are respectively equal to 0.9584 and 0.9437. However, as suggested by the radiologists performing the manual segmentation, few and sparse nonmatching pixels can hardly be considered as segmentation errors, being instead natural perceptual differences. Indeed, the loss of even small peripheral regions which could contain nodules cannot be tolerated. Based on these considerations, we classified as errors only those nonmatching connected regions larger than 13 pixels[4] and, according to this less strict criterium, we evaluated what we call "relaxed positive predictive value" and "relaxed sensitivity." We then say that an image is correctly segmented when both these measures rate at least 0.997.

Using this evaluation method, all the images in the JSRT database were correctly segmented; while in the Niguarda database errors were detected in six images, shown in Fig. 3 (i.e., 2.5% of the total number of images in this database). Table I lists, for the six images, the "relaxed positive predictive value," the "relaxed sensitivity," the "relaxed specificity," and the "relaxed accuracy." We also report in Table II the positive predictive value, the sensitivity, the specificity and the accuracy of the same images. In the last three rows of this table, we list the mean performance measures computed over the 247 images in the JSRT database, over the 234 images in the Niguarda database not containing errors, and over the whole set of 481 images, correctly segmented.

This performance was deemed good by the radiologists validating our system, especially because they reported the images in the Niguarda database as low-quality images, often characterized by low visibility near the costophrenic angles, the peripheral areas, the heart, and the diaphragm. In addition, many

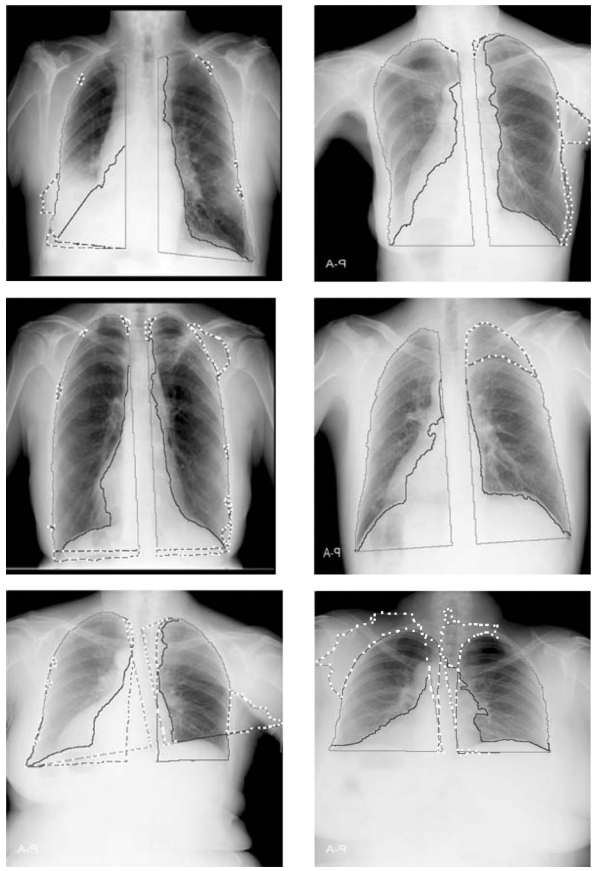


Fig. 3. Results on the Niguarda (more critical) database; the grey color shows the automatically segmented mask, the white dashed line shows the parts wrongly included or excluded. Top row: Image1, Image2. Central row: Image3, Image4. Bottom row: Image5, Image6.

TABLE I
"RELAXED POSITIVE PREDICTIVE VALUE," "RELAXED SENSITIVITY," "RELAXED SPECIFICITY," AND "RELAXED ACCURACY" FOR THE SIX IMAGES CONTAINING ERRORS

Image	*Rel. Pos. Pred.*	*Rel. Sens.*	*Rel. Spec.*	*Rel. Acc.*
Image1	0.9842	0.9875	0.9912	0.9899
Image2	0.9605	1.0000	0.9697	0.9826
Image3	0.9778	0.9623	0.9821	0.9732
Image4	1.0000	0.9468	1.0000	0.9771
Image5	0.8602	0.9037	0.9442	0.9331
Image6	0.8148	0.9669	0.9219	0.9337

[3]These measurements are generally used for the performance evaluation of classifiers.

[4]This is the pixel area of the digital circle whose radius is equal to the pixel radius of the smallest nodules to be detected in the image at 256 × 256 pixels.

TABLE II
POSITIVE PREDICTIVE VALUE, SENSITIVITY, SPECIFICITY, AND ACCURACY FOR THE SIX IMAGES CONTAINING ERRORS. LAST ROWS CONTAIN THE MEAN VALUES COMPUTED OVER THE 247 IMAGES IN THE JSRT DATABASE, OVER THE 234 CORRECTLY SEGMENTED IMAGES IN THE NIGUARDA DATABASE, AND OVER THE WHOLE SET OF 481 IMAGES CORRECTLY SEGMENTED

Image	*Pos . Pred.*	*Sensitivity*	*Specifcity*	*Accuracy*
Image1	0.9645	0.9493	0.9805	0.9693
Image2	0.8556	0.8857	0.8913	0.8889
Image3	0.9392	0.9305	0.9519	0.9424
Image4	1.0000	0.9245	1.0000	0.9674
Image5	0.6787	0.6875	0.8749	0.8229
Image6	0.7923	0.9443	0.9133	0.9214
247 JS RT	0.9925	0.9774	0.9937	0.9864
234 Niguarda	0.9908	0.9483	0.9923	0.9756
481 Images	0.9917	0.9632	0.9930	0.9811

images contain chests with structural abnormalities, the patients are often tilted or rotated, the position of the chest in the radiograph is not always at the center, and there is a lot of variation in the size of the thoraxes. The separation between the visible and the hidden areas, is generally precise (see Fig. 4).

The good performance of the method is also proved by comparing our results with those obtained through the method described in [63], which was applied to the JSRT database and reported to be one of the best performing. Since, in that case, the segmentation masks do not include the hidden areas, we have extended them near the spine and at the bottom, as described in [65]. By considering each nonmatching pixel as FP or FN, on the 247 images in the JSRT database our system obtained a mean positive predictive equal to 0.9925, a mean sensitivity equal to 0.9774 a mean specificity and accuracy, respectively, equal to 0.9937 and 0.9864, while the system presented in [63] obtained a mean positive predictive value equal to 0.9844, a mean sensitivity per image equal to 0.8701, and a mean specificity and accuracy, respectively, equal to 0.987 and 0.9303.

IV. ENHANCING THE CONSPICUITY OF THE NODULES

In this section, we describe the method used to enhance the conspicuity of nodules of various sizes and different contrast characteristics.

This step is common to all the schemes presented in the literature; they generally produce an enhanced image by means of background subtraction or an unsharp masking procedure, which is based on subtracting a smoothed version of the image from the original, or from an image where the high frequencies are highlighted (see [21], [22], [24], [37], and [43]). The result of course depends on the size of the operator used to produce the smoothed image: indeed it highlights only a limited range of frequencies, i.e., the details belonging to a limited scale. An example is the enhancement procedure used by the scheme presented in [20]. The authors use a smoothing operator of a certain size to enhance the visibility of the nodules. The candidate regions are then extracted by thresholding the result, and rules based on the shape are then used to discard some of them. We implemented the same method and applied it to the JSRT database: more than 30 nodules were missed, and the ones that were detected are the tumors with the biggest size and the lowest subtlety. Some others, with the same size but with spiculated texture characteristics were missed, since they are characterized by frequencies not detected by the filter. This is the reason to use both the multiscale enhancement method, described in this section, and the multiscale method for candidate nodule extraction, described in the next section. For efficiency, both methods are applied to the original radiographs down-sampled to 256×256 pixels.

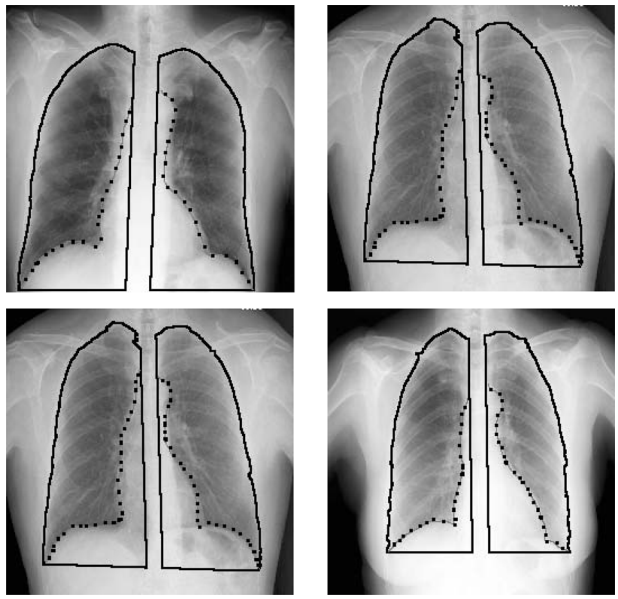


Fig. 4. Other results on the Niguarda database; the grey-black dotted line shows the automatically defined separation between the visible and the hidden lung areas.

The framework of a general multiscale scheme is often a procedure, parameterized by a scale parameter σ (see [68]–[70]), that can extract from the image the information belonging to a limited range of frequencies (scales) strictly related to σ itself. The final result is then created by integrating all the results obtained at the various levels of detail considered. An effective nodule detection scheme employing multiscale approaches is reported in [44]. We consider this method too complicate and present here a simpler technique to enhance and extract the nodule candidates.

Before applying the multiscale enhancement procedure, we apply a preprocessing step to remove some background information. This process has marked effect in the case of hidden lung areas, which are characterized by high gray levels and by low contrast due to the uniform texture of the superimposed structures (diaphragm, spine, heart). Some of this undesired information can be removed by a simple background-trend-correction technique; more precisely, from the original image, we subtracted a two-dimensional (2-D) surface $F(u,v)$ fitted to the gray levels in the image; $F(u,v)$ is a 2-D second-order polynomial whose coefficients are determined by using the least squares method (see Appendix I for a more detailed explanation). The new image $\tilde{I}(x,y)$ is calculated as

$$I_{Int}(x,y) = I(x,y) - F(x,y) \tag{5}$$

$$\tilde{I}(x,y) = I_{Int}(x,y) - \min(I_{Int}(x,y)) \tag{6}$$

where $I(x,y)$ is the pixel gray level and $\min(X)$ is the minimum value of the image X. The second operation is necessary to avoid negative gray level values.

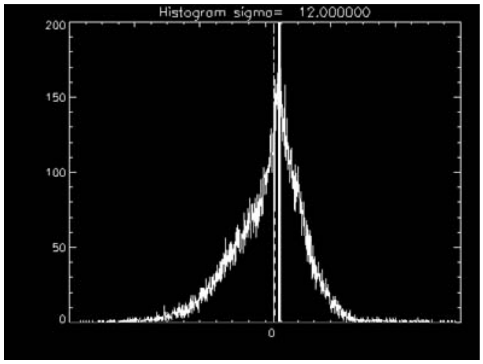


Fig. 5. Histogram of the difference image.

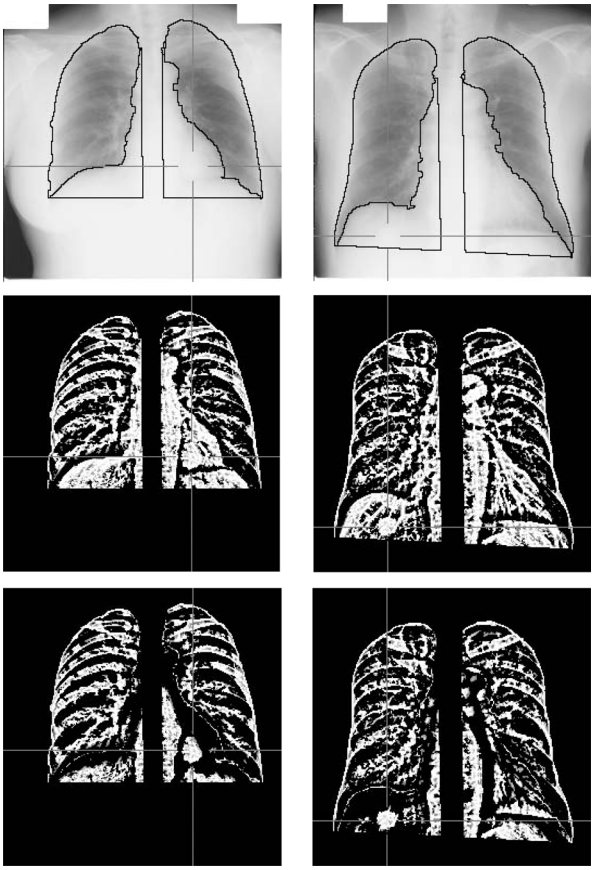


Fig. 6. Top row: Down-sampled images containing two subtle nodules. Middle row: Sum image obtained without preprocessing the input. Bottom row: Sum image obtained by the preprocessed input.

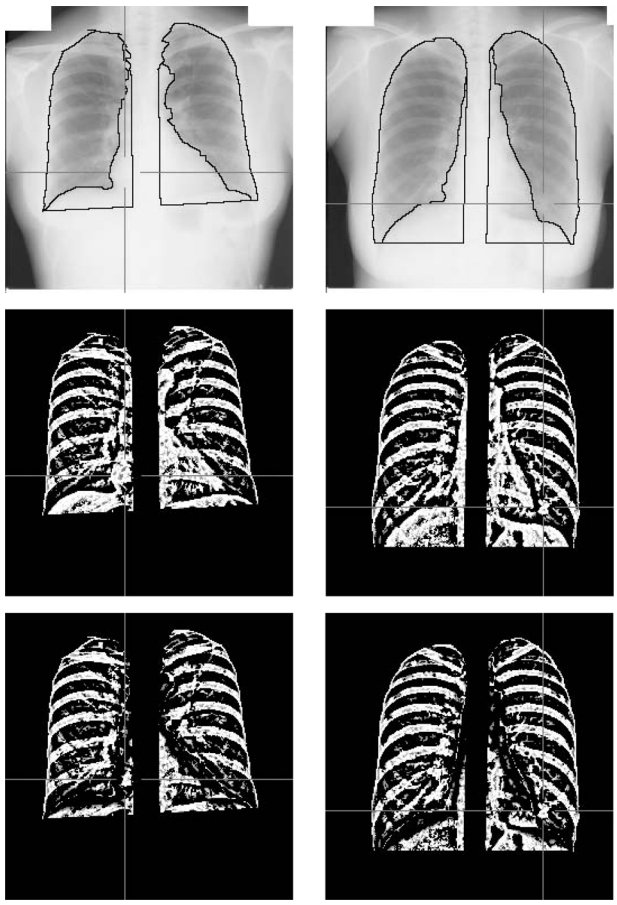


Fig. 7. Top row: Down-sampled images containing two extremely subtle nodules. Middle row: Sum image obtained without preprocessing the input. Bottom row: Sum Image obtained by the preprocessed input.

The image $\tilde{I}$ is further enhanced by histogram-equalization technique and the multiscale approach is applied to it; the aim is to enhance the visibility of nodules whose radius is in the [2.5 mm–17.0 mm] range. In the subsampled 256×256 image, this range corresponds to radius values in the set $\{2, 3, \ldots, 12\}$. Considering all 11 values in this set, we produce 11 smoothed versions of the image by convolving it with Gaussian filters whose standard deviation s takes values in $\{2, 3, \ldots, 12\}$. For each scale s, we then subtract from the original image its smoothed version, to get a difference image were the details visible at the scale s are enhanced. Since the gray level distribution in a nodule subimage can be approximated by a Gaussian distribution, the result of subtracting from a nodule subimage its smoothed version is usually an image with a positive peak in the central part of the nodule, and negative values in the neighborhood.

Moreover, the histogram of the difference image shows that most of the pixels take negative values, while on the set of positive values a peak can always be identified (see Fig. 5). We create a binary image by selecting all the pixels with a value greater than that of the peak. These pixels correspond to the highest frequencies, i.e., the details, that can be identified at the scale s. Summing up all the binary images obtained at the 11 different scales we get a final sum image, whose values are in the set $\{1, 2, \ldots, 11\}$.

Figs. 6 and 7 may be helpful in understanding the usefulness of preprocessing the image before applying the multiscale enhancement method. They show two original chest images in their top rows, the middle rows show the sum image obtained when using the original image as input, while the bottom rows contain the sum image produced when the preprocessed image is the input of the enhancement procedure. The reader may notice that in the sum images shown in the bottom rows the nodules appear much more visible; this gives visual evidence of the efficacy of applying the preprocessing step before the enhancement procedure.

In the next section a simple multiscale method is described, which processes the sum image in order to extract a set of potential nodules.

V. Extracting the Nodule Candidates

In the sum image, the nodules often appear as regions with circular shapes of various sizes, characterized by the highest gray levels at the center and surrounded by a much darker ring. Based on this observation, the image is processed to look for a measurement that helps in selecting the pixels that correspond to the centers of the nodules. To handle all possible nodule sizes, the procedure described below is repeated for each possible radius value ($r \in \{2,3,\ldots,12\}$) and all the results are then combined.

Having fixed the radius r, it is possible to calculate for each pixel $P = P(x,y)$ a coefficient $P_r(x,y)$ that measures the contrast between a circular region, centered in P with a radius r, and its surroundings; $P_r(x,y)$ is, thus, defined

$$P_r(x,y) = \mathrm{Mean}(\mathrm{Circle}_r(P(x,y))) - \mathrm{Mean}(\mathrm{Ring}_r(P(x,y))) \quad (7)$$

where $\mathrm{Mean}(X)$ is the mean of the pixel gray values inside a generic region X; $\mathrm{Circle}_r(P(x,y))$ is the circular region of radius r and centered in P; $Ring_r(P(x,y))$ is the region composed of the pixels in the two-pixel-thick ring around the $\mathrm{Circle}_r(P(x,y))$.

Note that the thickness of the ring is fixed at 2 for every radius. This is because what allows us to identify a circular region is the darker ring surrounding it, no matter what the thickness of the ring itself is.

To select the pixels that are potential nodule centers, the hidden areas are analyzed separately from the visible areas; indeed, two thresholds are defined on the two sets of their coefficients $\{P_r(x,y)\}$. Those values are automatically defined by means of the algorithm described in [71], which is based on the maximization of the entropy of the histogram of the coefficients. After thresholding, a binary image is created.

For each connected region in it, we calculate the area, the circularity,[5] and the greatest diagonal, D, of the minimum ellipse containing the region itself. We decided to discard a region either if it is too elongated (i.e., when D is bigger than $2.25r$ or when its circularity is lower than 0.6) or when it is too small (i.e., when its area is less than $\pi r^2/2$). Repeating the procedure for each radius value, we create 11 binary images; each image $B(r)$ contains a set of potential nodules of radius r. All these images must be combined to extract a final set of candidates. To this aim, a procedure is employed, which starts by considering as the initial candidate set, C_{set}, all the regions in $B(2)$; subsequently, for each consecutive radius r ($r \in \{3,\ldots,12\}$), the regions in $B(r)$ are checked to select new candidates to insert in C_{Set}, or to modify those already included.

More precisely, for each region $X \in B(r)$ ($r > 2$), if it does not overlap any element[6] $c \in C_{\mathrm{Set}}$, it is inserted in the candidate set. Otherwise, when the region X intersects a candidate c, the assumption is that X and c may be two different (and corresponding) representations of the same round pattern in the *Sum image*. Since a round pattern generally appears on consecutive scales with different shapes, c and X correspond to the same pattern if there is a region in $B(r-1)$ which intersects both X and c (this region, if present, is referred as the correspondent of X and c in $B(r-1)$). There are two cases.

- If no correspondent in $\boldsymbol{B(r-1)}$ is found, X is considered as a new candidate, different from c.
- If a correspondent in $\boldsymbol{B(r-1)}$ is found, the union of X and c ($u = X \cup c$), is first considered.
  If the biggest diagonal D of the minimum ellipse containing it is less or equal to $2.25r$, then u is taken as new candidate (and is inserted in C_{Set}); otherwise we compute the following values for X and c, respectively:

$$M_x = \frac{1}{|X|}\sum_{P\in X} P_r \text{ and } M_c = \frac{1}{|c|}\sum_{P\in c} P_r \quad (8)$$

  and we maintain in C_{Set} the one with the higher value.

Applying this extraction scheme to the 247 images in the JSRT database, only three nodules are missed (out of 154),[7] but the candidate set contains about 28 000 regions (i.e., more than 113 fp/image). Nevertheless, some of the false positives can be easily recognized by size or position. To detect and discard them, we compute five features for each candidate as follows.

- The normalized area is the region area divided by πR^2, where $R = 12$ is the maximum radius value.
- The normalized perimeter is the number of pixels in the region's external border divided by $2\pi R$, where $R = 12$ is the maximum radius value.
- The effective radius is the radius of the circle with an area equivalent to that of the region.
- The lung border overlapping (LBO) is calculated as the number of pixels in the region's external border that are outside the segmentation mask, divided by the total number of pixels in the external border itself; this quantity is bigger than zero only for those regions that are attached to the borders of the segmentation mask.
- The area fraction $\mathbf{Area_{Fr}}$): this quantity is based on the position of the region centroid[8], $\overline{C_m}$, and is used to identify those regions that are on the borders between the visible and the hidden lung areas. It is computed as the number of pixels in the region that are in the lung part where $\overline{C_m}$ is located, divided by the region area. This measure is less than 1.0 for those regions that are extended either in the hidden and in the visible areas, and it is approximately

[5]The circularity is defined as the fraction of the area of the region contained in the circle, with the same area and centered in the centroid, and the area of the circle itself (the same circularity measure has been used in [20]).

[6]Two regions are overlapping when the intersection of their pixels is not null.

[7]More precisely the missed nodules are in the radiographs JPCLN072.IMG (subtlety level 3, corresponding to "subtle nodule"), JPCLN138.IMG (subtlety level 1 corresponding to "extremely subtle nodule"), JPCLN145.IMG (subtlety level 1).

[8]The centroid $\overline{C_m}$ is calculated with the same equation used to calculate the center of mass, $\overline{C_{\mathrm{mass}}}$ by considering each pixel mass equal to 1.

$$\overline{C_{\mathrm{mass}}} = \frac{\sum_{i=1,N} m_i \overline{P_i}}{M} \quad (9)$$

where N is the number of pixels in the region, $\overline{P_i}$ is the position vector of the ith pixel and $M = \sum_{i=1,N} m_i$ is the total mass.

equal to 0.5 for those regions whose centroid is near the separation.

The computed values help to screen out those false positive regions whose size is too small or too large with respect to minimum and maximum nodule dimensions. Indeed, a candidate is discarded if:

- its normalized area is less than 0.1 or greater than 1.1;
- its normalized perimeter is less than 0.1 or greater than 1.5;
- its effective radius is less than 2 or greater than 15.

Our method basically detects those regions characterized by high contrast. This is the reason why it may wrongly detect some regions attached to the external lung border or located on the boundary between the visible and the hidden lung areas, where there is high contrast. Some of them can be discarded by means of their LBO and Area_{Fr} values. More precisely, we discard a candidate if:

- its Area_{Fr} is less than 0.6, since in this case the region is most probably a part of the border between the hidden and visible lung areas, and it has been wrongly detected as a nodule due to the high contrast that characterizes this area;
- its LBO is greater than 0.35, meaning that this region is attached to the borders for more than 1/3 of its perimeter.

These rules are effective and the fixed thresholds are not critical;[9] indeed, they are intuitive thresholds, which are not dependent on the database characteristics, but only on the possible nodule dimension. Their application to the initial set of about 28 000 candidates, allows us to significantly reduce its cardinality without missing any true positive region. The final candidate set is composed of 151 true positives and 18916 false positives, that is 76.58 fp/image.

After this deletion has been performed, a gray level image containing all the potential tumors, and called the regions image, is computed by assigning each pixel in each candidate region the value

$$G(x,y) = \max_{r\in\{2,\ldots,12\}} (P_r(x,y)) \tag{10}$$

and then scaling it to the range [0, 255] to obtain a gray level image. Fig. 8 shows two examples of a region image, containing a subtle (left) and an extremely subtle nodule (right).

These results have been compared to those of the extraction scheme reported in [45], that has been tested on the same database. The candidate set is extracted by the multiscale processing scheme, based on the scale-space theory introduced by Lindeberg [68]; this method aims at blob extraction and shape computation by means of (possibly nonlinear) combinations of Gaussian derivatives at multiple scales. This technique has been applied for several tasks due to its good results; nevertheless, it is applied only to the visible lung area, leading to a omission of 13 true positives, out of 154, even before candidate extraction. The result of the multiscale extraction scheme is a set of 33 000 candidates and a loss of another eight true positives.

[9]The threshold values were set by analyzing the feature distribution of a set of candidate nodules composed of 3500 randomly chosen false positives and 38 ($\approx 151/4$) true positives. The 38 true positives were randomly chosen according to their subtlety.

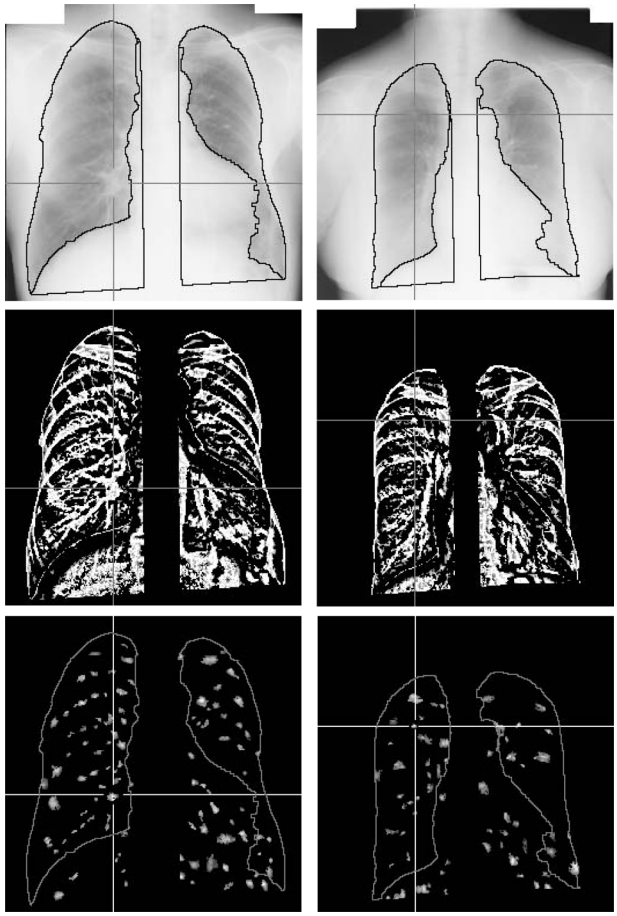


Fig. 8. Top row: Down-sampled image containing a subtle nodule (left) and an extremely subtle nodule (right). Middle row: Sum image. Bottom row: Regions Image.

To obtain an initial reduction in the candidates, the authors apply the KNN classifier, which reduces the candidates to 5028, but it misses another 15 true positives, in addition to the 21 already overlooked, for a total of 36 false negatives.

VI. Support Vector Machines for Candidates Classification

The high cardinality of the extracted candidate set led us to employ learning algorithms to discard the largest number of false positives. Indeed, most of the schemes presented in the literature extract an initial set of candidate nodules whose cardinality is too high. On the other hand, systems trying to extract a set with fewer candidates have been developed, but their high true positive miss rate has lead to approaches that leave the task of reduction to proper classifiers. The most effective methods presented in the literature employ learning systems, such as feed-forward artificial neural networks with different architectures (e.g., [25]–[27], [30], [33], [39], [42], [44], [46]). Some more successful approaches (e.g., [45]) were based on two consecutive classification steps to first reduce the large number of candidates and then extract the final subset.

In the last years, support vector machines (SVMs) have been widely investigated and used in a lot of different fields and for various classification tasks, due to their good performances.

In the field of medical image processing, several authors (e.g., [72]–[76]) have presented CAD schemes which employ SVMs for candidate classification. Our previous experiments using a rule-based system [49], and learning algorithms such as neural networks [77] and support vector machines [78], both trained with different parameters and input features, showed that support vector machines produce the most robust results. In these experiments, the best performance was obtained with a data set made up of 16 features describing the shape and the gray level characteristics of the candidates.

Nevertheless, the results obtained were still not satisfactory, due to the high number of false positive candidates left after classification (we achieved a sensitivity equal to 0.7 and about 16 fp/image). With the aim of increasing classification performance, we calculated a set of 160 features including the 16 features previously mentioned. We then applied the feature selection technique based on the univariate Golub's statistics [79]. Considering a dichotomic problem the Golub's statistic of the ith feature is formulated as

$$\mathrm{Golub}(i) = \frac{\mu_i^+ - \mu_i^-}{\sigma_i^+ + \sigma_i^-} \tag{11}$$

where i refers to the ith feature, μ_i^+ (σ_i^+) and μ_i^- (σ_i^-) are the means (and the standard deviations) of the ith feature of the positive and negative examples. Basically, this value assesses whether the means of two groups are statistically different from each other; the value $\mathrm{Golub}(i)$ for the ith feature, can therefore be interpreted as a measurement of the separation between the two classes, if the data were represented only with the ith feature itself. Precisely, the $\mathrm{Golub}(i)$ values computed for the 160 features are in the range $[-0.2755, 0.5697]$.

To create an initial data set, we decided to select the features with a positive value of $\mathrm{Golub}(i)$ greater than 0.175. Since there were 18 features above this value we symmetrically selected the 18 features with the most negative values (their $\mathrm{Golub}(i)$ value is less than -0.085). We, thus, created a data set made up of 36 features (the 36 features data set).

To test the performance of the learning system when the number of selected features decreases, we also tested SVMs trained on a subset (18 features data set), which was created by selecting the nine features with the most positive values ($\mathrm{Golub}(i)$ greater than 0.36), and the nine features with the most negative value ($\mathrm{Golub}(i)$ less than -0.16).

The 160 features computed to create these data sets are sketched in section Section VI-A and they are described in details in appendix II. The experimental setup is described in Section VI-B and the results of the classification tasks, performed on the two data sets, are reported in Section VII.

The overall method, including feature computation, was implemented with IDL, an interpreted language. When executed on a Pentium 4 (2.6 GHz) with 512 MB RAM, it takes 45 s per image. This time can be reduced to approximately 4.5 s if the algorithm is developed in a compiled language.

A. Data Sets

For each candidate region, we calculated the following five sets of features (see appendix II for their detailed description).

1) **19** features describe the shape and position of the region;
2) **16** features describe the gray level distribution of the region pixels in the original X-ray image (down-sampled to 1024×1024 pixels), in the *Regions Image*, and in the *Sum Image*;
3) **6** features are related to the estimates of the radius value computed for the candidate region;
4) **11** features are computed by applying *LoG* filters, at 11 different scales, to the subimage containing the candidate region;
5) **108** features are computed by applying the oriented first derivative of Gaussian filters, at six different scales and eight different directions, to the subimage containing the candidate region.

Applying the feature selection technique previously described we selected 36 features to form the 36 features data set and its subset, i.e., the 18 features data set (see Appendix II where the list of the features composing each data set is reported). The data vectors in both feature sets were normalized to 0 mean and unitary standard deviation, before being used for training and testing.

Since the number of true positives extracted (151) is much lower than the number of false positives (18916), both the 36 features and the 18 features Data Sets are very unbalanced; this explains the positive-enriched data sets that were built for training and testing the SVMs. For each data set, we separately considered the true positive (simply referred to as positive in the hereafter) and the false positive (negative) examples, and we randomly split the available positive data into 136 examples for training and 15 examples for testing, according to a train/test ratio of 9/1.[10] From the set of negative data, we extracted without replacement a number of negative examples equal to 30 times the number of positive data obtaining, respectively, $136 \times 30 = 4080$ negative examples for the training set and $15 \times 30 = 450$ negative examples for the test set. To perform an unbiased evaluation, we repeated the above process 10 times, obtaining 10 pairs of training and test sets.

To test system performance when the number of positive examples used for training decreases, we ran other tests using a train/test ratio equal to 7/3, i.e., 106 examples for training and the remaining 45 for testing. In this case, we had $106 \times 30 = 3180$ negative examples in the training set and $45 \times 30 = 1350$ negative examples in the test set. This process was repeated 10 times, obtaining 10 pairs of training and test sets.

Using the two data sets (the 36 features data set and the 18 features data sets), and the two ratios, 9/1 and 7/3, for splitting the positive examples into the train/test sets, we ran four experiments for each SVM, whose results are reported in the following section (see Section VII).

B. Experimental Setup

In this context, we need to obtain high sensitivity in order to detect all the positive examples, without a significant loss in specificity, because, from a medical point of view, it is crucial to detect all positive examples, but, at the same time, we need to significantly reduce the number of negatives. For these reasons,

[10]This ratio was chosen to have a significant number of true positive cases during the training process.

all the results were judged on the basis of their sensitivity and specificity.

With SVMs, by lowering the decision threshold, we can increase sensitivity at the possible expense of a decreased specificity. Hence, to better understand the behavior of the classifiers, we performed a receiver operating characteristic (ROC/FROC) analysis to jointly and succinctly evaluate the sensitivity and specificity of the trained SVMs.

For each data set, we tested nine polynomial and nine Gaussian SVMs; the nine different polynomial kernels were created by setting the polynomial degree $d \in \{2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6\}$, while the nine Gaussian kernels were created by varying the width, σ, of the Gaussian kernels between 10 and 1400 ($\sigma \in \{10,100,500,600,700,800,900,1200,1400\}$). Note that linear SVMs were not tested because our previous experiments [77] had shown their poor performance.

Traditional SVMs employ a 0/1 loss function, equally weighing errors on both positive and negative examples. This may be a problem when using a small number of positive examples compared to negative ones, since we might obtain too low a sensitivity [80]. Moreover, when dealing with medical problems, equally weighted errors may be inappropriate, because the cost of misclassification of the positive (diseased) patients is usually higher than misclassification of negative (healthy) patients.

For this reason, in this context, we applied a cost-sensitive approach. In particular, in the framework of the SVM optimization problem, two regularization parameters, C_+ and C_-, were introduced (see [81]) to be able to adjust the cost of misclassification of false positives versus false negatives. Following this approach, in the experiments presented here, we set $C_- = C$ and $C_+ = C \times C_f$, where C and C_f are the regularization parameter and the cost-factor, respectively.

More precisely, for both the Gaussian and the polynomial kernels, we ran experiments where C_f was set to 2, 5, 10, 20, 50, 100, 200, 300, 400 (so that training errors on positive examples outweigh errors on negative examples), and we varied the regularization parameter C from 0.001 to 200 (using the values 0.001, 0.005, 0.01, 0.05, 0.1, 0.5, 0.75, 1, 5, 10, 15, 25, 50, 100, 200). Thus, for each of the nine different Gaussian or polynomial kernels, we tested 135 different SVM models (obtaining a total of 1215 SVM models for each kernel).

We used the ten pairs of training and test sets to train and test each SVM model; sensitivity and specificity were calculated with respect to the test sets.

VII. Results

We considered, for each data set, all the pairs of sensitivity and specificity obtained on the test sets with the different polynomial and Gaussian SVM models.

We report below the best results obtained, for both the data sets and the two train/test ratios.

A. Results Obtained With the 36 Features Data Set

1) Results Obtained With the 36 Features Data Set and 9/1 Train/Test Ratio: In the case of the 9/1 train/test ratio, the performance of the polynomial kernels is slightly better than that of the Gussian kernels, and both kernels types yield their best results with low values of C ($C \leq 15$) and a value of the cost factor, C_f, often equal to 20.

TABLE III
Results Obtained With SVMs Trained on the 36 Features Data Set, When the Train/Test Ratio Is Equal to 9/1. Results Are Sorted by Sensitivity

SVM kernel	Sens.	Spec.	fp/ image
Poly $d = 2$, $C = 0.01$, $C_f = 20$	0.7143	0.9738	2.0
Poly $d = 2$, $C = 0.05$, $C_f = 20$	0.7143	0.9714	2.19
Gauss $\sigma = 100$, $C = 1$, $C_f = 20$	0.7143	0.9667	2.55
Gauss $\sigma = 100$, $C = 0.5$, $C_f = 20$	0.7857	0.9524	3.65
Gauss $\sigma = 600$,$C = 7.5$, $C_f = 50$	0.7857	0.9452	4.19
Gauss $\sigma = 900$,$C = 15$, $C_f = 20$	0.8571	0.9357	4.92
Poly $d = 2$, $C = 0.05$, $C_f = 10$	0.8571	0.9333	5.1
Gauss $\sigma = 100$, $C = 0.5$, $C_f = 20$	0.9286	0.9214	6.02
Poly $d = 2$, $C = 0.005$, $C_f = 50$	0.9286	0.9096	6.93
Gauss $\sigma = 100$, $C = 1$, $C_f = 20$	1	0.9024	7.47
Gauss $\sigma = 100$,$C = 0.5$, $C_f = 20$	1	0.8976	7.84
Poly $d = 2$, $C = 0.05$, $C_f = 20$	1	0.8762	9.48

The polynomial kernels yielded better results when the degree of the polynomial is low (it is generally $d = 2, 3$), while Gaussian kernels generally yielded better results when the σ takes high values ($\sigma \geq 100$). Under these conditions, we always obtain no more than 3 fp/image when the sensitivity is greater than 0.7, no more than 4 fp/image when the sensitivity is about 0.78, about 5 fp/image when it increases to values greater than 0.85, and no more than 6 or 8 fp/image when the sensitivity increases to about 0.92 and 0.1, respectively.

Table III shows some of the best results obtained; the first column reports the characteristics of the SVM: the kernel type and the corresponding regularization, and kernel parameters. In the second and third columns, the sensitivity and the specificity are represented. In the last column, the mean number of remaining false positives per image is shown.

Some examples of ROC curves obtained with four of the models described in Table III are shown in Fig. 9. In the same image we also show two examples of free-response receiver operating characteristic (FROC) curves [82], obtained with a Gaussian and a polynomial kernel. Each image shows four curves that describe the behavior of the SVM model for four different test sets.

2) Results Obtained With the 36 Features Data Set and 7/3 Train/Test Ratio: In the case of 7/3 train/test ratio the classification performance of the SVM models decreases, meaning that the number of positive cases used for training influences the learning capability of the SVM.

In this case, Gaussian and polynomial SVMs yielded similar performance. In general, we obtained about 5 fp/image when the sensitivity was more than 0.7. When sensitivity increases to about 0.75 we obtain 6 or 7 fp/image. For sensitivity approximately equal to 0.9 and 0.95, we get about 13 and 20 fp/image, respectively. More precisely, the Gaussian kernels perform better with high values of σ ($\sigma \geq 500$), low values of C ($C \leq 25$), and a cost factor generally equal to 20. Polynomial SVMs perform better when the polynomial degree is equal to 2, the cost factor is equal to 20 or 50, and C assumes truly low values. Some detailed results are reported in Table IV.

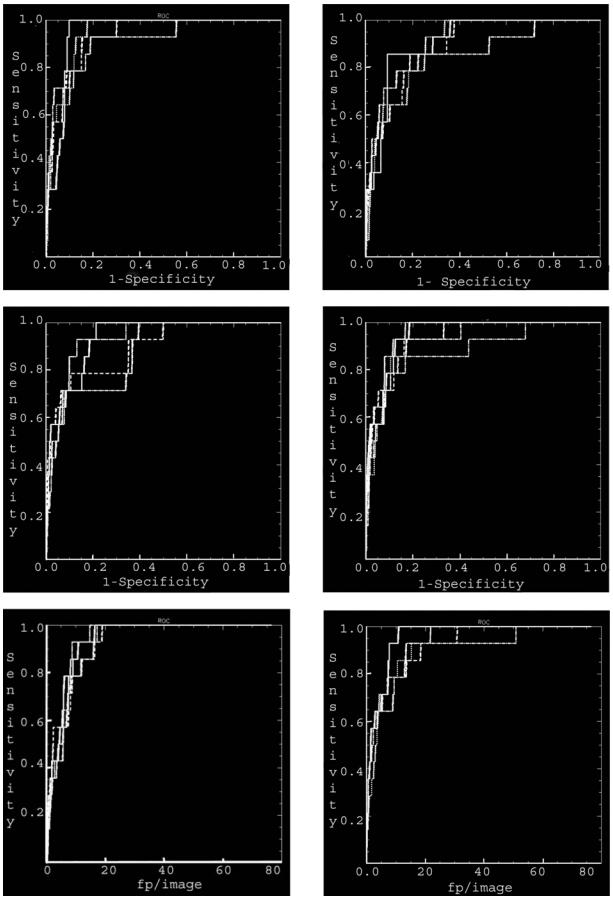


Fig. 9. ROC curves obtained with the 36 features data set and train/test ratio equal to 9/1. Top row: ROC curves obtained with a Gaussian kernel with $\sigma = 100$ and $C = 1$ and cost factor equal to 20 (left) and $\sigma = 100$ and $C = 15$ and cost factor equal to 50 (right). Central row: ROC curves obtained with a polynomial kernel with $d = 2$ and $C = 0.5$ and cost factor equal to 20 (left) and a polynomial kernel with $d = 2$ and $C = 0.05$ and cost factor equal to 10 (right). Bottom row: FROC curves obtained with a Gaussian kernel with $\sigma = 100$ and $C = 0.1$ and cost factor equal to 50 (left) and a polynomial kernel with $d = 2$ and $C = 0.5$ and cost factor equal to 5 (right). Each image contains four curves that show the behavior of the SVM model for four different test sets drawn from the 36 features data set.

TABLE IV
RESULTS OBTAINED WITH SVMS TRAINED ON THE *36 FEATURES* DATA SET, WHEN THE TRAIN/TEST RATIO IS EQUAL TO 7/3. RESULTS ARE SORTED BY SENSITIVITY

SVM kernel	Sens.	Spec.	fp/image
Poly $d = 2$, $C = 0.05$, $C_f = 20$	0.7057	0.9326	5.16
Gauss $\sigma = 700$, $C = 25$, $C_f = 20$	0.7055	0.9303	5.34
Poly $d = 2$, $C = 0.005$, $C_f = 50$	0.7564	0.9129	6.67
Gauss $\sigma = 500$, $C = 7.5$, $C_f = 20$	0.7512	0.9053	7.25
Gauss $\sigma = 600$, $C = 0.01$, $C_f = 50$	0.9091	0.8273	13.22
Poly $d = 2$, $C = 0.005$, $C_f = 20$	0.9091	0.8227	13.57
Poly $d = 2$, $C = 0.1$, $C_f = 50$	0.9545	0.7295	20.71
Gauss $\sigma = 600$, $C = 0.01$, $C_f = 50$	0.9545	0.722	21.29

B. Results Obtained on the 18 Features Data Set

The results obtained with this data set are comparable to those obtained with the 36 features data set, meaning that the discarded features have too low a discriminating power; indeed, their deletion does not decrease the learning capability of the SVM.

TABLE V
RESULTS OBTAINED WITH SVMS TRAINED ON THE *18 FEATURES* DATA SET, WHEN THE TRAIN/TEST RATIO IS EQUAL TO 9/1. RESULTS ARE SORTED BY SENSITIVITY

SVM kernel	Sens.	Spec.	fp/image
Poly $d = 3$, $C = 0.1$, $C_f = 10$	0.7143	0.981	1.45
Poly $d = 4$, $C = 0.05$, $C_f = 10$	0.7143	0.981	1.46
Gauss $\sigma = 1100$, $C = 25$, $C_f = 50$	0.7143	0.9738	2.00
Poly $d = 3$, $C = 0.1$, $C_f = 20$	0.7857	0.9667	2.55
Poly $d = 3$, $C = 0.1$, $C_f = 20$	0.7857	0.9667	2.55
Gauss $\sigma = 10$, $C = 0.1$, $C_f = 20$	0.7857	0.9643	2.73
Poly $d = 2$, $C = 0.001$, $C_f = 100$	0.8571	0.9595	3.10
Poly $d = 4$, $C = 0.05$, $C_f = 10$	0.8571	0.9429	4.37
Gauss $\sigma = 10$, $C = 0.05$, $C_f = 10$	0.8571	0.9405	4.55
Gauss $\sigma = 10$, $C = 0.1$, $C_f = 20$	0.9286	0.9071	7.11
Poly $d = 3$, $C = 0.01$, $C_f = 50$	0.9286	0.9	7.65
Poly $d = 2$, $C = 0.001$, $C_f = 100$	0.9286	0.8976	7.84
Poly $d = 3$, $C = 0.01$, $C_f = 50$	1	0.8976	7.84
Poly $d = 3$, $C = 0.01$, $C_f = 10$	1	0.8867	8.67

1) Results Obtained With the 18 Features Data Set and 9/1 Train/Test Ratio: Again in this case, polynomial kernels generally performed better than Gaussian kernels because they yielded lower values of fp/image with the same sensitivity. Precisely, with polynomial kernels we get about 1.5 fp/image for sensitivity approximately equal to 0.71, no more than 2.8 fp/image for sensitivity around 0.78, no more than 4.7 fp/image when the sensitivity increases to 0.85, no more than 8 and 9 fp/image when the sensitivity increases to approximately 0.9 and 1.0, respectively. With the Gaussian kernels we generally obtain about 2.5 fp/image for sensitivity approximately equal to 0.71, no more than 3.5 fp/image for sensitivity around 0.78, and no more than 6 fp/image when the sensitivity increases to 0.85. For the highest sensitivities (approximately equal to 1.0), we get about 11 fp/image.

Table V shows that such performance is obtained with the polynomial kernels when all the parameters take low values ($C \leq 0.5$, $d = 3, 4$ and $C_f \leq 20$). Gaussian SVMs performed better when trained with low values of both C ($C \leq 0.5$) and cost factor ($C_f \leq 50$), while σ generally takes large values ($\sigma \geq 500$). Some examples of ROC curves obtained with two of the models described in Table V are shown in the top row of Fig. 10. In the central row, two FROC curves obtained with a Gaussian and a polynomial kernel are shown, while in the bottom row, modified ROC curves are shown for two different polynomial SVM models; these curves report sensitivity versus fp/image.

2) Results Obtained With the 18features Data Set and 7/3 Train/Test Ratio: As with the *36Features Data Set*, when the SVMs are trained with a train/test ratio of 7/3 performance decreases. We generally obtain 6 or 7 fp/image when sensitivity is around 0.7, and this number grows to approximately 8, 15 and 20 fp/image when sensitivity increases to about 0.75, 0.91, and 0.95, respectively. As in the previously described experiments, both polynomial and Gaussian SVMs yielded better results with a low value of the cost factor ($C_f = 20, 50$) and a low value of C ($C \leq 10$). Gaussian kernels yielded better results when σ was large ($\sigma \geq 500$), while polynomial kernels perform better when the degree of the polynomial is equal to 2, 3, or 4, as in the

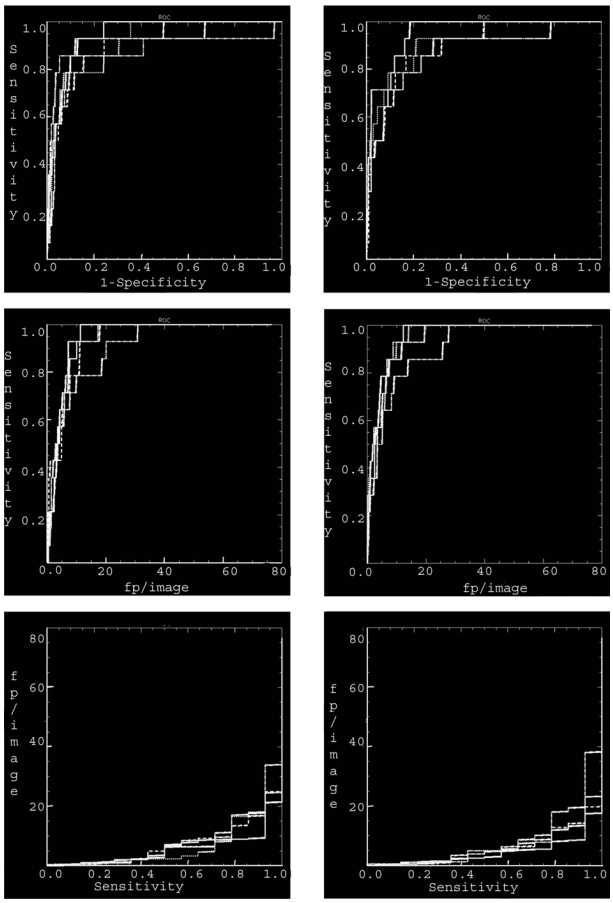


Fig. 10. Plots obtained with the 18 features data set and train/test ratio equal to 9/1. Top row: ROC curves obtained with Gaussian kernels with $\sigma = 10$, $C = 0.1$ and cost factor equal to 20 (left) and polynomial kernel with polynomial degree $d = 4$, $C = 0.05$ and cost factor equal to 10 (right). Central row: FROC curves obtained with Gaussian kernels with $\sigma = 10$, $C = 0.1$ and cost factor equal to 50 (left) and polynomial kernel with polynomial degree $d = 5$, $C = 0.005$ and cost factor equal to 20 (right). Bottom row: Plots of sensitivity versus fp/image. They are both obtained with a 9/1 train/test ratio and polynomial kernels; the degree of the polynomial is $d = 3$ and the cost factor is equal to 50, meanwhile $C = 0.01$ (left) and $C = 0.005$ (right).

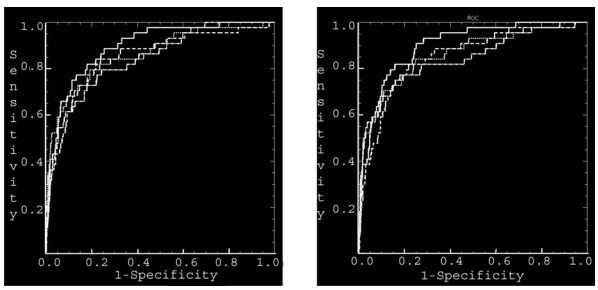


Fig. 11. ROC curves obtained with the 18 features data set and train/test ratio equal to 7/3 and cost factor equal to 20. Polynomial kernels of degree $d = 2$ and $C = 0.05$ (left) and Gaussian kernel with $\sigma = 500$ and $C = 7.5$ (right).

previous experiments. Some examples of *ROC* curves obtained are shown in Fig. 11.

As mentioned in Section VI, our previous candidate classification experiments employed a different nodule candidate set, whose elements were classified by means of SVM models trained on a data set consisting of 16 features. This set of features yielded good performance when used as input for a rule-based system. To obtain proper comparison, we computed the 16 features on the current candidate set. Using the data set thus obtained (the 16 Features Data Set), we performed the same learning experiments with all the SVM models and both the 9/1 and 7/3 train/test ratio. Table VI list some of the best results obtained when the train/test ratio is equal to 9/1. These results are worse than those presented above, proving that feature selection techniques greatly improve classifier performance. The results obtained with a 7/3 train/test ratio are not shown, since they also were worse than the results previously presented.

TABLE VI
RESULTS OBTAINED WITH SVMS TRAINED ON THE *16 FEATURES* DATA SET, AND A 9/1 TRAIN/TEST RATIO

SVM kernel	Sens.	Spec.	fp/img
Poly $d = 4, C = 0.01$, $C_f = 50$	0.7857	0.9452	4.19
Gauss $\sigma = 100$, $C = 2$, $C_f = 100$	0.7857	0.9167	6.38
Poly $d = 4, C = 0.1$, $C_f = 50$	0.8571	0.9262	5.65
Gauss $\sigma = 100$, $C = 2$, $C_f = 100$	0.8571	0.8714	9.85
Poly $d = 3, C = 0.001$, $C_f = 100$	0.9286	0.8905	8.38
Gauss $\sigma = 100$, $C = 2, C_f = 50$	0.9286	0.8881	8.56
Poly $d = 4, C = 0.001$, $C_f = 100$	1.0	0.8252	13.39
Poly $d = 2, C = 0.1$, $C_f = 100$	1.0	0.8143	14.22

VIII. CONCLUSION

This paper presents a completely automatic system for lung nodule detection. The system was tested on the JSRT database, containing 247 X-ray images and 154 nodules. At first, a set of candidate nodules is extracted by means of a simple extraction scheme which is applied to the automatically segmented lung area. Due to the high number of false positives extracted, a set of 160 features was calculated and a feature selection technique was applied to select the best features. Cost-sensitive polynomial and Gaussian support vector machines have been tested and trained with two data sets that differ in size, i.e., the number of features composing the data set. Comparison of the results showed that polynomial kernels generally yielded better performance; moreover the two data sets offered comparable results, meaning that the larger data set comprises more features whose discriminating power is too low, so they do not increase the classifier learning capability. With the best models, we obtain ≈ 1.5 and ≈ 2.5 fp/image for sensitivity equal to ≈ 0.71 and ≈ 0.78, respectively. When sensitivity increases to ≈ 0.85, we get ≈ 4.5 fp/image. When sensitivity takes its highest values (≈ 0.92 and 1.0) no more than ≈ 7 or ≈ 8 fp/image are left.

Analyzing the nodules lost when the sensitivity is fixed at 0.71, we observe that they are either located in highly vascularized areas, or hidden behind ribs and clavicles. Due to these complex backgrounds, the extraction scheme does not produce a precise segmentation of the nodule areas and, as a consequence, the features computed in the next step are less accurate making the SVM classification task more difficult. However, we observe that no nodules classified as "obvious" (first subtlety level) are lost, while the lost nodules are almost equally distributed among the other four subtlety classes. This suggests that the difficulties encountered by radiologists are not the same as those found by the computer program, thus enforcing the idea that an automatic nodule detection method might provide a useful "second

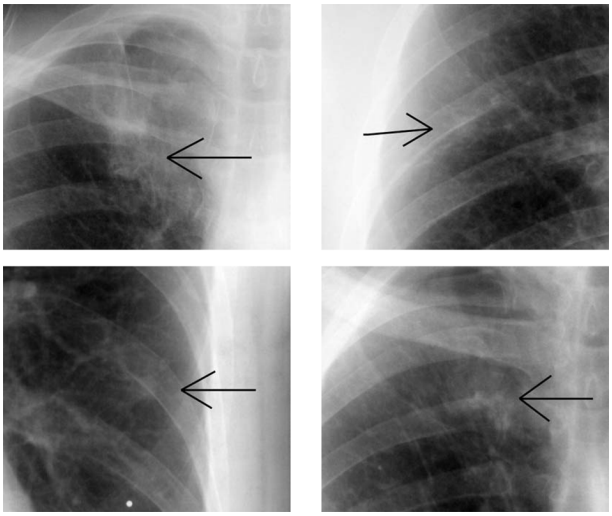


Fig. 12. Some examples of nodules lost. Top row: "relatively obvious" and "subtle" nodules; files JPCLN022.IMG and JPCLN066.IMG. Second row: "Very subtle" and "extremely subtle" nodules; files JPCLN107.IMG and JPCLN143.IMG.

opinion." Fig. 12 shows some examples of nodules lost when sensitivity is 0.71.

Comparison of our results to those obtained with a data set made up of 16 features and used in our previous candidate-classification experiments proves that the feature selection technique greatly improves classification performance, and makes the system useful for clinical practise. Note that SVMs trained on a data set where the train/test ratio is equal to 9/1 always perform better than those trained on a data set with the same features and a 7/3 train/test ratio. This suggests that the number of positive examples used for training has great impact on the results. The system could therefore obtain even better performance when trained on a larger database containing a greater number of nodules.

Appendix I
Polynomial Fit of a Two-Dimensional Surface

A grey level image $I(x,y)$, with a $N \times M$ pixel dimension, can be approximated by a 2-D surface, $F(x,y)$, having the polynomial form

$$F(x,y) = \sum_{d=0}^{n} \sum_{d'=0}^{n'} A_{d,d'}\, x^d\, y^{d'} \tag{12}$$

which corresponds to an nth degree polynomial expansion in x and an n'th degree polynomial expansion in y. For each point, (x,y), $n \cdot n'$ expansion coefficients, $A_{d,d'}$, are determined by least-squares fitting, that is by minimizing the sum of square differences (SSD):

$$\mathrm{SSD} = \sum_{x,y} \left(I(x,y) - F(x,y)\right)^2. \tag{13}$$

In our implementation, we set $n = n' = 2$.

Appendix II
Input Features

The 19 features describing the candidate position and its shape are as follows.

- The six features described in Section V:
  - normalized area ($\mathrm{Norm}_{\mathrm{Area}}$);
  - normalized perimeter ($\mathrm{Norm}_{\mathrm{Perim}}$);
  - effective radius (Eff_R);
  - lung border overlapping (LBO);
  - area fraction Area_{Fr});
  - circularity (Circ).
- The perimeter of the region divided by the perimeter of the circle with radius equal to the effective radius (Eff_P).
- The two coordinates of the center of mass, (C_M), of the region; it is calculated with the (9), where the mass of each pixel is its value in the *Regions Image*.
- the two coordinates of the centroid of the region (C_B);
- the two coordinates of the centroid of the region (CNorm_B) expressed with respect to a coordinate system centered on the lung where the region is located, and normalized with respect to the height and width to the same lung;
- The Euclidean distance (Dist_C) between the center of mass and the centroid of the region.
- A binary feature (C_{Pos}), which has a value equal to 1 if the centroid of the region is in the visible lung areas, and 0 otherwise.
- The fraction between the two dimensions of the minimum bounding box containing the region ($\mathrm{MinBB}_{\mathrm{frac}}$).
- The fraction between the two dimensions of the maximum bounding box ($\mathrm{MaxBB}_{\mathrm{frac}}$).
- The two features that express the compactness[11] of the region with respect to its minimum ($\mathrm{Comp}_{\mathrm{MINBB}}$) and its maximum ($\mathrm{Comp}_{\mathrm{MAXBB}}$) bounding box.

The **16** features computed to describe the gray level distribution of the candidate region pixels are the mean, the standard deviation, the maximum value and the contrast[12] of the pixels values in the regions image, in the sum image and in the X-ray image down-sampled to 1024×1024 pixels. These features will be referred as $\mathrm{Mean}_{\mathrm{RegImg\backslash SumImg\backslash Img}}$, $\mathrm{StdDev}_{\mathrm{RegImg\backslash SumImg\backslash Img}}$, $\mathrm{Max}_{\mathrm{RegImg\backslash SumImg\backslash Img}}$, and $\mathrm{Contr}_{\mathrm{RegImg\backslash SumImg\backslash Img}}$.

The remaining four features are the normalized gray level mean and the normalized contrast of the region pixels in the image and in the regions image, respectively ($\mathrm{Mean}^{\mathrm{Norm}}_{\mathrm{Img\backslash RegImg}}$, $\mathrm{Contr}^{\mathrm{Norm}}_{\mathrm{Img\backslash RegImg}}$); normalization is performed with respect to the maximum gray level, and the maximum contrast found in the hidden or in the visible areas (in the Image and in the Regions Image, respectively), depending on the location of the region centroid.

The 6 features related to the radius estimates associated with the candidate region are based on the coefficients $P_r(x,y)$ computed for each pixel and each radius value. To obtain three radius

[11]It is computed as the fraction of the area of the region and the area of its bounding box.

[12]It is calculated as the difference between the maximum and the minimum values of the pixel gray levels.

features, a most eligible radius, $\mathrm{rad}(x,y)$, is first computed for each pixel $P = P(x,y)$. This is done by considering all the P_r calculated for that pixel, so that $\mathrm{rad}(x,y)$ is equal to

$$\mathrm{rad}(x,y) = \mathrm{Argmax}_{i\in\{2,12\}}(P_i(x,y)). \tag{14}$$

Two radius features, R_1 and R_2, associated with region X are then equal to

$$R_1(X) = \mathrm{Mean}_{(x,y)\in X}(\mathrm{rad}(x,y)) \tag{15}$$

$$R_2(X) = \max_{(x,y)\in X}(\mathrm{rad}(x,y)). \tag{16}$$

Another radius estimate R_3 is computed by considering, for each different value of r, the sum

$$\mathrm{Sum}(X,r) = \sum_{(x,y)\in X} P_r(x,y) \tag{17}$$

R_3 is calculated according to

$$R_3(X) = \mathrm{Argmax}_{i\in\{2,12\}}(\mathrm{Sum}(X,r)). \tag{18}$$

Other two radius estimates, FirstR and LastR, are the first and last radius value for which a representation of the region has been detected. The last feature related to the radius estimate is the radius standard deviation, StdR, which was calculated as

$$\mathrm{StdR(X)} = \mathrm{StdDev}_{(x,y)\in X}(\mathrm{rad}(x,y)) \tag{19}$$

where StdDev(Z) is the standard deviation of the values in the set Z.

The last **119** features, described below, are calculated on a sub-image extracted from the image down-sampled to 1024×1024 pixels and containing the candidate region. All the candidate subimages are extracted with a size of 101×101.[13] The decision to compute the features on the whole candidate subimage is due to the fact that it contains information related both to the candidate region and to its surroundings; this added information may increase the features' discriminating power.

Before feature extraction, each subimage was processed in order to remove background information. To this end, we have subtracted from the subimage itself a second order polynomial function, which was fitted to the subimage itself, according to (5) (see Section IV and Appendix 12). This operation may enhance some unnecessary and really fine detail that belong to the vessels or the borders of the ribs. To remove them, we apply a five-pixel median filter and we convolve the image with a three-pixel smoothing filter, whose coefficients are those of the Gaussian function with 0.5 standard deviation.[14]

[13]This size has been chosen since the largest nodule in the image down-sampled to 1024×1024 pixels has a pixel radius equal to 48 pixels. Extracting the subimage of 101×101 pixels allows us to include the largest nodule and some external information.

[14]The sizes of both the median and the smoothing filters have been chosen to be little in order to avoid losing important details in the candidate subimage.

The pixel values are then normalized into the range [0., 500.], since 500 is the mean contrast value found in all the candidate subimages. The candidates subimage thus processed is then used to compute the remaining features.

11 features are calculated by means of the LoG filters at 11 different scales, in the range $\{8,\ldots,48\}$;[15] they are extracted by taking, for each scale, the mean values computed for the centroid pixel and its eight neighbors.

108 features are computed by convolving the subimage with a bank of first derivatives of Gaussian filters oriented along eight directions and at six different scales. The six scales are chosen so that the support of the oriented filter varies in the range $\{8,\ldots,48\}$ of the radius values evaluated in the image down-sampled to 1024×1024 pixels; more precisely $s = \{8,16,24,32,40,48\}$. For each scale s and for each orientation d we calculated the mean filter output computed for the pixels in the candidate region ($\mathrm{GMean_{Reg}}(s,d)$), thus obtaining eight values for each scale. To evaluate the response variation of the filter along the eight directions, we also computed the standard deviation ($\mathrm{StdG_{Re0g}}(s)$) of the eight mean values extracted at each scale.

The same feature-extraction procedure is applied by considering the values of all the pixels in the subimage; thus, obtaining for each scale s another set of nine features ($\mathrm{GMean_{SubImg}}(s,d),\ d = \{1,\ldots,8\}$ and $\mathrm{StdG_{SubImg}}(s)$).

Applying the feature selection technique previously described, we selected 36 features to form the 36 features data set and its subset, i.e., the 18 features data set. More precisely, the 36 features data set is made up of the following:

1) **13** features that describe the shape and position of the region, which are: $\mathrm{Norm_{Area}}$, $\mathrm{Norm_{Perim}}$, Eff_R, Area_{Fr}, Eff_P, C_M, C_B, CNorm_B, $\mathrm{Comp_{MINBB}}$, $\mathrm{Comp_{MAXBB}}$;
2) **7** features that describe the gray level distribution of the region pixels in the original image (down-sampled to 1024×1024 pixels), in the regions image, and in the sum image; they are: $\mathrm{StdDev_{Img}}$, $\mathrm{Contr_{Img}}$, $\mathrm{Contr_{Img}^{Norm}}$, $\mathrm{Mean_{RegImg}}$, $\mathrm{Contr_{RegImg}}$, $\mathrm{Max_{RegImg}}$, $\mathrm{Mean_{SumImg}}$;
3) all the **6** features related to the radius estimate associated with the candidate region;
4) **10** features computed by applying the first derivative of Gaussian filters at different orientations and scales; they are: $\mathrm{StdG_{Reg}}(s),\ s = \{8,16,24,32,40\}$ and $\mathrm{StdG_{SubImg}}(s),\ s = \{8,16,24,32,40\}$.

Note that none of the 11 features computed by applying the LoG filters have been selected.

The 18 features data set is made up of:

1) **7** features that describe the shape and position of the region, which are: $\mathrm{Norm_{Area}}$, $\mathrm{Norm_{Perim}}$, Eff_R, C_B, CNorm_B;
2) **4** features that describe the gray level distribution of the region pixels in the image (down-sampled to 1024×1024 pixels) and in the regions image; they are: $\mathrm{Contr_{Img}^{Norm}}$, $\mathrm{Mean_{RegImg}}$, $\mathrm{Contr_{RegImg}}$, $\mathrm{Max_{RegImg}}$;
3) **3** features related to the radius estimate associated with the candidate region, which are: FirstR, LastR, and StdR;

[15]The scale used for the LoG filter is four times the 11 radius values used by the enhancement and the extraction scheme; this choice is due to the fact that we are no longer working on the image down-sampled to 256×256 pixels but we are extracting the features from the image down-sampled to 1024×1024 pixels.

4) **4** features computed by applying the first derivative of Gaussian filters: $\mathrm{StdG}_{\mathrm{Reg}}(s)$, $s = \{16, 24\}$ and $\mathrm{StdG}_{\mathrm{SubImg}}(s)$, $s = \{24, 32\}$.

The 16 features used in our previous experiments are

1) **11** features that describe the shape and position of the region which are $\mathrm{Norm}_{\mathrm{Area}}$, $\mathrm{Norm}_{\mathrm{Perim}}$, Eff_R, ***LBO***, Area_{FR}, Circ, C_M, C_{Pos}, $\mathrm{MinBB}_{\mathrm{frac}}$, $\mathrm{MaxBB}_{\mathrm{frac}}$;
2) **3** features that describe the gray level distribution of the region pixels in the original radiograph and in the regions image; they are: $\mathrm{Mean}_{\mathrm{Img}}$, $\mathrm{Mean}_{\mathrm{RegImg}}$, $\mathrm{Max}_{\mathrm{RegImg}}$;
3) **2** features related to the radius estimate associated with the candidate region, which are: R_1, R_3.

ACKNOWLEDGMENT

The authors would like to thank Dr. A. Vanzulli, Head of the Radiology Department, Niguarda Ca'Granda Hospital, Milan, for suggesting us the problem and for his help in evaluating the system performance; Prof. G. Valentini for his helpful discussion and eager suggestions; the reviewers for their precious comments and advice, which enabled them to vastly improve their paper.

REFERENCES

[1] G. P. Murphy, W. Lawrence Jr., and R. E. Lenhard Jr., *Clinical Oncology*. Washington, DC: Amer. Cancer Soc., 1995.
[2] J. D. Steele, W. P. Kleitsch, J. E. Dunn, and P. Buell, "Survival in males with bronchogenic carcinomas resected as asymptomatic solitary pulmonary nodules," *Annotation Thoracic Surg.*, vol. 2, pp. 368–376, 1966.
[3] J. W. Vance, C. A. Good, C. H. Hodgson, J. W. Kirklin, and R. P. Gage, "The solitary pulmonary lesion due to bronchogenic carcinoma: A 3-year follow-up study on 94 surgically treated patients," *Disease Chest*, vol. 36, pp. 231–237, 1959.
[4] P. E. Buell, "The importance of tumor size in prognosis for resected bronchogenic carcinoma," *J. Oncological Surg.*, vol. 3, pp. 539–551, 1971.
[5] T. Heelan, B. J. Flehinger, M. R. Melamed, M. B. Zaman, W. B. Perchick, J. F. Caravelli, and N. Martini, "Non-small-cell lung cancer: Result of the New York screening program," *Radiology*, vol. 151, pp. 289–293, 1984.
[6] H. Zhao, S. C. B. Lo, M. T. Freedman, and Y. Wang, "Enhanced lung cancer detection in temporal subtraction chest radiography using directional filtering techniques," in *Proc. SPIE*, 2002, vol. 4684.
[7] E. Samei, M. J. Flynn, W. Eyler, and E. Peterson, "Subtle lung nodules: Influence of local anatomic variations on detection," *Radiology*, vol. 228, pp. 76–84, 2003.
[8] L. W. Guiss and P. Kuenstler, "A retrospective view of survey photofluorocarbons of persons with lung cancer," *Cancer*, vol. 13, pp. 91–95, 1960.
[9] J. R. Muhm, W. E. Miller, R. S. Fontana, D. R. Sanderson, and M. A. Uhlenhopp, "Lung cancer detected during a screening program using fourth-month chest radiographs," *Radiology*, vol. 19, pp. 609–615, 1983.
[10] F. P. Stitik, "Screening for cancer: Chest radiology," *Miller AB*, pp. 163–191, 1985.
[11] J. Forrest and P. Friedman, "Radiologic errors in patient with lung cancer," *West J. Med.*, vol. 134, pp. 485–490, 1981.
[12] J. H. M. Austin, B. M. Romeny, and L. S. Goldsmith, "Missed bronchogenic carcinoma: Radiographic findings in 27 patients with apotentially resectable lesion evident in retrospect," *Radiology*, vol. 182, pp. 115–122, 1992.
[13] G. Gavelli and E. Giampalma, "Sensitivity and specificity of chest X-ray screening for lung cancer," in *Proc. Int. Conf. Prevention Early Diagnosis Lung Cancer*, Varese, Italy, 1998, pp. 103–108.
[14] T. Matsumoto, H. Yoshimura, K. Doi, M. L. Giger, A. Kano, H. MacMahon, M. Abe, and S. Montner, "Potential usefulness of computerized nodule detection in screening programs for lung cancer," *Investigative Radiol.*, vol. 27, pp. 471–475, 1992.
[15] T. Kobayashi, X.-W. Xu, H. MacMahon, C. Metz, and K. Doi, "Effect of a computer-aided diagnosis scheme on radiologists' performance in detection of lung nodules on radiographs," *Radiology*, vol. 199, pp. 843–848, 1996.
[16] H. MacMahon, K. Doi, R. Engelmann, F. M. Behlen, K. R. Hoffman, T. Ishida, C. Roe, and C. E. Metz, "Computer aided diagnosis of pulmonary nodules: Results of a large-scale observer study," *Radiology*, vol. 23, pp. 723–726, 1999.
[17] S. Kakeda, J. Moriya, H. Sato, T. Aoki, H. Watanabe, H. Nakata, N. Oda, S. Katsuragawa, K. Yamamoto, and K. Doi, "Improved detection of lung nodules on chest radiographs using a commercial computer-aided diagnosis system," *Amer. J. Roentgenol.*, vol. 182, pp. 505–510, 2003.
[18] P. Sankar and J. A. Slansky, "Gestalt guided heuristic boundary follower for X-ray images of lung nodules," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 4, pp. 326–331, 1982.
[19] W. Lampeter and J. Wadtke, "Computerized search of chest radiographs for nodules," *Investigative Radiol.*, vol. 21, pp. 384–390, 1986.
[20] M. Giger, K. Doi, and H. MacMahon, "Image feature analysis and computer-aided diagnosis in digital radiography: Automated detection of nodules in peripheral lung fields," *Med. Phys.*, vol. 15, pp. 158–166, 1988.
[21] M. L. Giger, N. Ahn, K. Doi, H. MacMahon, and C. E. Metz, "Computerized detection of pulmonary nodules in digital chest images: Use of morphological filters in reducing false positive detections," *Med. Phys.*, vol. 17, pp. 861–865, 1990.
[22] M. Giger, K. Doi, H. MacMahon, C. Metz, and F. Yin, "Pulmonary nodules: Computer-aided detection in digital chest images," *Radiographics*, vol. 10, pp. 41–51, 1990.
[23] K. Suzuki and N. Inakoa, "Development of a computer aided detection system for lung cancer diagnosis," *Proc. SPIE*, vol. 1652, pp. 567–571, 1992.
[24] H. Yoshimura, M. Giger, K. Doi, H. MacMahon, and S. Montner, "Computerized scheme for the detection of pulmonary nodules: A non linear filtering technique," *Investigative Radiol.*, vol. 27, pp. 124–127, 1992.
[25] S.-C. Lo, M. Freedman, J.-S. Lin, and S. Mun, "Automatic lung nodule detection using profile matching and backpropagation neural network techniques," *J. Digital Imag.*, vol. 1, pp. 48–54, 1993.
[26] M. F. McNitt-Gray, J. W. Sayre, H. K. Huang, and M. Razavi, "A pattern classification approach to segmentation of chest radiographs," *Proc. SPIE*, vol. 1898, pp. 160–170, 1993.
[27] Y. Wu, K. Doi, M. Giger, C. Metz, and W. Zhang, "Reduction of false positives in computerized detection of lung nodules in chest radiographs using artificial neural networks, discriminant analysis, and a rule-based scheme," *J. Digital Imag.*, vol. 7, no. 4, pp. 196–207, 1994.
[28] A. Kano, K. Doi, H. MacMahon, D. D. Hassel, and M. Giger, "Digital image subtraction of temporally sequential chest images for detection of interval change," *Med. Phys.*, vol. 21, pp. 453–461, 1994.
[29] H. Yoshida, X. Xu, K. Doi, and M. Giger, "Computer-aided diagnosis (CAD) scheme for detecting pulmonary nodules using wavelet transforms," *Proc. SPIE*, vol. 2434, pp. 621–626, 1995.
[30] S.-C. Lo, S. L. Lou, J.-S. Lin, M. Freedman, M. Chien, and S. Mun, "Artificial convolution neural network techniques and applications for lung nodule detection," *IEEE Trans. Med. Imag.*, vol. 14, no. 4, pp. 711–718, Dec. 1995.
[31] J. S. Lin *et al.*, "Differentiation between nodules and end-on vessels using a convolution neural network architecture," *J. Digital Imag.*, vol. 8, pp. 132–141, 1995.
[32] C. Floyd *et al.*, "Diffuse nodular lung disease on chest radiographs: A pilot study of characterization by fractal dimension," *Amer. J. Roentgenol.*, vol. 167, pp. 1185–1187, 1996.
[33] J. S. Lin, S. C. Lo, M. Freedman, and S. Mun, "Reduction of false positives in lung nodule detection using a two-level neural classification," *IEEE Trans. Med. Imag.*, vol. 15, no. 2, pp. 206–217, Apr. 1996.
[34] C. Vyborny, "The AAPM/RSNA physics tutorial for residents: Image quality and the clinical radiographic examination," *Radiographics*, vol. 17, pp. 479–498, 1997.
[35] X.-W. Xu *et al.*, "Development of an improved CAD scheme for automated detection of lung nodules in digital chest images," *Med. Phys.*, vol. 24, pp. 1395–1403, 1997.
[36] F. MAo *et al.*, "Fragmentary window filtering for multiscale lung nodule detection: Preliminary study," *Acad. Radiol.*, vol. 5, pp. 306–311, 1998.
[37] X.-W. Xu, S. Katsuragawa, A. Ashizawa, H. MacMahon, and K. Doi, "Analysis of image features of histograms of edge gradient for false positive reduction in lung nodule detection in chest radiographs," in *Proc. SPIE*, 1998, vol. 3338, pp. 318–326.

[38] J. M. Carreira *et al.*, "Computer-aided diagnosis:Automatic detection of lung nodules," *Med. Phys.*, vol. 25, pp. 1998–2006, 1998.

[39] M. Penedo, M. Carreira, A. Mosquera, and D. Cabello, "Computer aided diagnosis: A neural network based approach to lung nodule detection," *IEEE Transaction on Medical Imaging*, vol. 17, no. 6, pp. 872–880, Dec. 1998.

[40] D. Catarious Jr., A. Baydush, and C. Floyd Jr., "Initial development of a computer aided diagnosis tool for solitary pulmonary nodules," in *Proc. SPIE*, 2001, vol. 4322, pp. 710–717.

[41] Q. Li, S. Katsuragawa, and K. Doi, "Computer-aided diagnostic scheme for lung nodule detection in digital chest radiographs by use of a multiple-template matching technique," *Med. Phys.*, vol. 28, pp. 2070–2076, 2001.

[42] J. Wei, Y. Hagihara, A. Shimizu, and H. Kobatake, "Optimal image feature set for detecting lung nodules on chest x-rays images," in *CARS 2002*, 2002.

[43] K. Bilgin and H. Yoshida, "Computerized detection of pulmonary nodules in chest radiographs based on morphological features and wavelet snake model," *Med. Image Anal.*, vol. 6, pp. 431–447, 2002.

[44] G. Coppini, S. Diciotti, M. Falchini, N. Villari, and G. Valli, "Neural networks for computer-aided diagnosis: Detection of lung nodules in chest radiograms," *IEEE Trans. Inf. Technol. Biomed.*, vol. 7, no. 4, pp. 344–357, Dec. 2003.

[45] A. Schilham, B. Van Ginneken, and M. Loog, "Multi-scale nodule detection in chest radiographs," in *Proc. Medical Image Computing and Computer-Assisted Intervention, MICCAI*. New York: Springer, 2003, vol. 2878, Lecture Notes on Computer Science, pp. 602–609.

[46] K. Suzuki, J. Shiraishi, H. Abe, H. MacMahon, and K. Doi, "False-positive reduction in computer-aided diagnostic scheme for detecting nodules in chest radiographs by means of massive training artificial neural network," *Acad. Radiol.*, vol. 12, pp. 191–201, 2005.

[47] B. van Ginneken, B. M. t. H. Romeny, and M. A. Viergever, "Computer-aided diagnosis in chest radiography: A survey," *IEEE Trans. Med. Imag.*, vol. 20, no. 12, pp. 1228–1241, Dec. 2001.

[48] J. Shiraishi, S. Katsuragawa, J. Ikezoe, T. Matsumoto, T. Kobayashi, K. Komatsu, M. Matsui, H. Fujita, Y. Kodera, and K. Doi, "Development of a digital image database for chest radiographs with and without a lung nodule: Receiver operating characteristic analysis of radiologists' detection of pulmonary nodules," *Amer. J. Roentgenol.*, vol. 174, pp. 71–74, 2000.

[49] P. Campadelli and E. Casiraghi, "A nodule detection system for postero-anterior chest radiographs," in *Proc. Modelling, Computation Optimization Inf. Syst. Management Sci. (MCO2004)*, Metz, France, 2004.

[50] D. Cheng and M. Goldberg, "An algorithm for segmenting chest radiographs," *Proc. SPIE*, pp. 261–268, 1998.

[51] A. Hasegawa, S.-C. Lo, M. T. Freedman, and S. K. Mun, "Convolution neural network based detection of lung structure," in *Proc. SPIE*, 1994, vol. 2167, pp. 654–662.

[52] E. Pietka, "Lung segmentation in digital chest radiographs," *J. Digital Imag.*, vol. 7, pp. 79–84, 1994.

[53] M. F. McNitt-Gray, H. K. Huang, and J. W. Sayre, "Feature selection in the pattern classification problem of digital chest radiographs segmentation," *IEEE Trans. Med. Imag.*, vol. 14, no. 3, pp. 537–547, Sep. 1995.

[54] X.-W. Xu and K. Doi, "Image feature analysis for computer aided diagnosis: Accurate determination of ribcage boundaries chest radiographs," *Med. Phys.*, vol. 22, pp. 617–626, 1995.

[55] ——, "Image feature analysis for computer aided diagnosis: Accurate determination of right and left hemidiaphragm edges and delineation of lung field in chest radiographs," *Med. Phys.*, vol. 23, pp. 1616–1624, 1996.

[56] J. Duryea and J. M. Boone, "A fully automatic algorithm for the segmentation of lung fields in digital chest radiographic images," *Med. Phys.*, vol. 22, pp. 183–191, 1995.

[57] S. G. Armato, M. Giger, and H. MacMahon, "Automated lung segmentation in digitized posteroanterior chest radiographs," *Academic Radiol.*, vol. 5, pp. 245–255, 1998.

[58] M. S. Brown, L. S. Wilson, B. D. Doust, R. W. Gill, and C. Sun, "Knowledge-based method for segmentation and analysis of lung boundaries in chest x-rays images," *Comput. Med. Imag. Graph.*, vol. 22, pp. 463–477, 1998.

[59] F. M. Carrascal, J. M. Carreira, M. Souto, P. G. Tahoces, L. Gomez, and J. J. Vidal, "Automatic calculation of total lung capacity from automatically traced lung boundaries in postero-anterior and lateral digital chest radiographs," *Med. Phys.*, vol. 25, pp. 1118–1131, 1998.

[60] O. Tsuji, M. T. Freedman, and S. K. Mun, "Automated segmentation of anatomic regions in chest radiographs using an adaptive-sized hybrid neural network," *Med. Phys.*, vol. 25, pp. 998–1007, 1998.

[61] N. F. Vittitoe, R. Vargas-Voracek, and C. E. Floyd Jr., "Identification of lung regions in chest radiographs using Markov random field modeling," *Med. Phys.*, vol. 25, pp. 976–985, 1998.

[62] ——, "Markov random field modeling in posteroanterior chest radiograph segmentation," *Med. Phys.*, vol. 26, pp. 1670–1677, 1999.

[63] B. van Ginneken and B. M. t. H. Romeny, "Automatic segmentation of lung fields in chest radiographs," *Med. Phys.*, vol. 27, pp. 2445–2455, 2000.

[64] B. van Ginneken, "Computer-aided diagnosis in chest radiographs," Ph.D. dissertation, Univ. Utrecht, The Netherlands, 2001.

[65] P. Campadelli and E. Casiraghi, "Lung field segmentation in digital postero-anterior chest radiographs," in *Proceedings of 3th International Conference on Advances in Pattern Recognition (ICAPR2005)*. New York: Springer-Verlag, 2005, vol. LNCS 3687:II, Lecture Notes in Computer Science by , pp. 736–745.

[66] T. Uchiyama and M. A. Arbib, "Color image segmentation using competitive learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 16, no. 12, pp. 1197–1206, Dec. 1994.

[67] F. J. Estrada and A. D. Jepson, "Quantitative evaluation of a novel image segmentation algorithm," in *Proc. Comput. Vision Pattern Recognition (CVPR2005)*, Jun. 20–25, 2005, vol. 2, pp. 1132–1139.

[68] T. Lindeberg, "Scale-space theory in computer vision," in *The Kluwer International Series in Engineering and Computer Science*. Norwell, MA: Kluwer, 1994.

[69] ——, "Scale-space: A framework for handling image structures at multiple scales," in *Proc. CERN School Comput.*, The Netherlands, Sep. 8–21, 1996.

[70] ——, "Feature detection with automatic scale selection," *Int. J. Comput. Vision*, vol. 30, no. 2, pp. 79–116, 1998.

[71] J. N. Kapur, P. K. Sahoo, and A. K. C. Woong, "A new method for gray level picture thresholding using the entropy of the histogram," *Comput. Vision Graph. Image Process.*, vol. 29, pp. 273–285, 1985.

[72] Z. Jun, L. Xiaomao, L. Jianguo, P. Fuyuan, T. Jinwen, W. Ying, Z. Wenjun, and X. Mingxing, "SVM-based ultrasonic medicine image diagnosis," *Proc. SPIE Med. Image Acquisition Process.*, vol. 4549, pp. 92–95, 2001.

[73] S. Xuejun, Q. Wei, and S. Dansheng, "Three-class classification in computer-aided diagnosis of breast cancer by support vector machine," *Proc. SPIE Med. Imag.*, vol. 5370, pp. 999–1007, 2004.

[74] S. S. Mohamed and M. M. A. Salama, "Computer-aided diagnosis for prostate cancer using support vector machine," *Proc. SPIE Med. Imag.*, vol. 5744, pp. 898–906, 2005.

[75] D. Glotsos, J. Tohka, P. Ravazoula, D. Cavouras, and G. Nikiforidis, "Automated diagnosis of brain tumours astrocytomas using probabilistic neural network clustering and support vector machines," *Int. J. Neural Syst.*, vol. 15, pp. 1–11, 2005.

[76] A. K. Jerebko, J. D. Malley, M. Franaszek, and R. M. Summers, "Support vector machines committee classification method for computer-aided polyp detection in CT colonography," *Academic Radiol.*, vol. 12, no. 4, pp. 479–486, 2005.

[77] E. Casiraghi, "A computer aided diagnosis system for lung nodules detection in postero anterior chest radiographs," Ph.D. dissertation, Università degli Studi di Milano, Milano, Italy, 2005.

[78] P. Campadelli, E. Casiraghi, and G. Valentini, "Support vector machines for candidate nodules classification," *Neurocomputing*, vol. 68, pp. 281–288, 2005.

[79] T. R. Golub *et al.*, "Molecular classification of cancer: Class discovery and class prediction by gene expression monitoring," *Science*, vol. 286, pp. 531–537, 1999.

[80] P. Campadelli, E. Casiraghi, and G. Valentini, "Lung nodules detection and classification," in *Proc. IEEE Int. Conf. Image Processing (ICIP2005)*, Sep. 2005, pp. 1117–20.

[81] K. Morik, P. Brockhausen, and T. Joachims, "Combining statistical learning with a knowledge-based approach—A case study in intensive care monitoring," in *Proc. Int. Conf. Machine Learning (ICML)*, 1999, pp. 268–277.

[82] D. P. Chakraborty and L. H. L. Winter, "Free-response methodology: Alternate analysis and a new observer-performance experiment," *Radiology*, vol. 174, no. 3, pp. 873–881, 1990.